

C# & .NET Deep Dive

Senior .NET Developer Interview Guide

Part 8 | System Design Case Studies | Senior Q&A

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How to use this guide

- Use each answer as a speaking framework: clarify, design, justify trade-offs, and connect the decision to a production scenario.
- Do not memorize word-for-word. Understand the flow and explain it naturally in your own project language.
- Practice the follow-up questions. Senior interviews test depth, failure handling, scale, security, and operational readiness.

Interview Question

What to expect

Strong Answer

How to answer

Real Project Lens

How to sound senior

What is included

01. System design interview approach	10. SQL vs NoSQL selection	19. Large file import processing
02. Requirement clarification and assumptions	11. Data modeling, indexing and partitioning	20. Notification service
03. Capacity estimation and SLAs	12. Caching and cache invalidation	21. Payment or order workflow
04. High-level architecture and data flow	13. Rate limiting and traffic control	22. Background job scheduling
05. Monolith vs modular monolith vs microservices	14. Authentication, authorization and secrets	23. Audit trail and compliance
06. API contract design and versioning	15. Resiliency patterns	24. Multi-tenant SaaS design
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1. How to approach a system design interview

Interview Question: You are asked to design a new enterprise system. How do you begin?

Strong Answer: I begin by clarifying the business goal, users, core workflows, scale, latency, consistency, availability, security, and failure expectations. I state assumptions, propose a simple high-level design, and then deepen the discussion around APIs, data, communication, resiliency, observability, and trade-offs. I do not start by naming cloud services or drawing many boxes.

A senior answer should follow a clear sequence

Stage	What I explain
1. Clarify	Users, use cases, constraints, scale, SLAs and assumptions
2. Design	Entry points, services, data stores, messaging and request flow
3. Protect	Security, validation, rate limiting and failure isolation
4. Operate	Logging, metrics, tracing, alerts, deployment and support
5. Close	Trade-offs, bottlenecks, risks and future improvements

Good interview line

"I will first confirm the requirements and scale. Then I will present the simplest design that meets them, and I will add complexity only where a requirement justifies it."

2. Requirement clarification and assumptions

Interview Question: What questions do you ask before designing the system?

Strong Answer: I ask who uses the system, the most important user journeys, expected traffic and data volume, read/write ratio, acceptable latency, availability target, consistency requirement, retention, compliance, integrations, and expected growth. When information is missing, I make an explicit assumption and confirm that the interviewer is comfortable with it.

Questions that make the answer sound senior

- Is the workflow real-time, asynchronous, batch-oriented, or a combination?
- What must never be duplicated, lost, exposed, or processed out of order?
- What is the source of truth, and which views can be eventually consistent?
- What failure behavior is acceptable to the business?
- What are the peak load, retention period, and regulatory constraints?

Real project lens

For a logistics import platform, I would clarify file size, record count, validation rules, processing window, retry expectations, user-visible status, audit requirements, and whether partial success is allowed. Those answers drive the architecture more than the technology preference.

3. Capacity estimation and SLAs

Interview Question: How do you estimate scale during a system design interview?

Strong Answer: I use rough calculations, not false precision. I estimate peak requests per second, average payload size, daily data growth, retention, read/write ratio, concurrency, and downstream limits. I translate business expectations into measurable targets such as p95 latency, availability, recovery time, recovery point, throughput, and queue backlog limits.

Example estimation framework

Area	Example assumption	Design impact
Traffic	2 million requests/day, 10x peak	Autoscaling, load balancing, rate limits
Data	5 KB/request, 365-day retention	Storage growth, partitioning, archival
Latency	p95 under 300 ms	Caching, indexing, async offloading
Availability	99.9% monthly	Redundancy, health checks, failover

Senior-level point

Capacity estimates are used to identify likely bottlenecks and validate the design. They are not meant to impress the interviewer with exact arithmetic.

4. High-level architecture and data flow

Interview Question: How do you explain a high-level architecture diagram?

Strong Answer: I explain the request from left to right: client, edge or gateway, application services, data stores, asynchronous components, external integrations, and observability. For each component I state its responsibility, data ownership, communication style, scaling model, and failure behavior.

Example request flow



What the interviewer should hear

- Where authentication and authorization are enforced.
- Which service owns each business capability and database.
- Which operations are synchronous and which are asynchronous.
- How failures, retries, duplicate messages, and timeouts are handled.
- How a support team traces one request across the system.

Avoid

Do not describe only the happy path. A senior design must explain what happens when the database is slow, a dependency is unavailable, a message is duplicated, or a deployment is partially unhealthy.

5. Monolith vs modular monolith vs microservices

Interview Question: When would you choose a monolith, modular monolith, or microservices?

Strong Answer: I choose based on domain complexity, team boundaries, independent scaling, deployment frequency, data ownership, and operational maturity. A well-structured modular monolith is often the safest starting point. Microservices are justified when independent ownership, scaling, release cadence, isolation, or technology boundaries create clear value.

Trade-off comparison

Option	Strength	Risk
Monolith	Simple deployment and transactions	Tight coupling as the system grows
Modular monolith	Clear boundaries with lower operational cost	Requires discipline to preserve modules
Microservices	Independent scale, ownership and deployment	Distributed data and operational complexity

Senior-level answer

I would not split services by technical layer such as one controller service and one database service. I would split by business capability and ownership, and only after the boundaries are understood.

6. API contract design and versioning

Interview Question: What do you consider while designing an enterprise API contract?

Strong Answer: I design around stable business resources and use cases. The contract defines request and response models, validation, status codes, authentication, authorization, pagination, filtering, idempotency, concurrency, error format, correlation ID, and versioning. I avoid exposing internal database entities directly.

Contract checklist

- Use predictable resource naming and HTTP semantics.
- Return a consistent error envelope with code, message, field errors, correlation ID, and safe details.
- Use pagination and limits for collections; never return unbounded data.
- Use backward-compatible additive changes where possible.
- Version only when a breaking contract change is unavoidable.

Real project example

For a long-running import, POST /imports returns 202 Accepted with a job ID and status URL. GET /imports/{id} returns progress, counts, warnings, and failure details. This avoids keeping an HTTP request open for the entire processing window.

7. Idempotency and duplicate request handling

Interview Question: How do you prevent duplicate processing when clients retry?

Strong Answer: For operations that create financial, order, job, or external side effects, I require an idempotency key or stable business key. I persist the key with the request fingerprint and final response inside the same transaction as the business state. A retry with the same key returns the previous result instead of repeating the operation.

Typical design

- Validate that the same key is not reused with a different payload.
- Use a unique constraint to make concurrent duplicate requests safe.
- Store processing, completed, and failed states with an expiry policy.
- Make message consumers idempotent because brokers can redeliver.
- Use deduplication as a safety mechanism, not as a replacement for correct transactions.

Real project lens

If a payment provider times out after accepting a charge, blindly retrying can double-charge the customer. I keep the same provider reference and idempotency key, query final status, and reconcile before creating a new attempt.

8. Synchronous vs asynchronous communication

Interview Question: When do you use synchronous APIs and when do you use messaging?

Strong Answer: I use synchronous communication when the caller needs an immediate response and the dependency is part of the user-facing decision. I use asynchronous messaging for long-running work, integration, buffering, fan-out, retries, and workflows where temporary delay is acceptable. I avoid converting every call to messaging because that increases operational and consistency complexity.

Decision guide

Use synchronous when	Use asynchronous when
Immediate validation or query result is required	Work can complete later
Simple request-response ownership	Traffic spikes need buffering
Failure must be returned to caller	Consumers need independent retries
Low fan-out	Multiple subscribers react to an event

Senior-level point

Even with asynchronous processing, the API should return a durable acknowledgment only after the job or event is safely stored.

9. Messaging, delivery guarantees and DLQ

Interview Question: How do you design reliable event-driven processing?

Strong Answer: I assume at-least-once delivery unless the platform and end-to-end workflow prove otherwise. Producers publish durable events, consumers process idempotently, offsets or acknowledgments are committed only after successful handling, and poison messages are isolated in a dead-letter queue with enough context to diagnose and replay them.

Important senior-level terms

Concern	Design response
Duplicate delivery	Idempotent consumer and unique business key
Ordering	Partition by an ordering key only where required
Poison message	Bounded retries, DLQ and replay process
Producer consistency	Transactional Outbox or equivalent
Schema change	Versioned, backward-compatible event contracts

Good interview line

"Exactly once" is usually a business outcome created through idempotency, atomic state changes, and reconciliation, not only a broker configuration.

10. SQL vs NoSQL selection

Interview Question: How do you choose between a relational and a NoSQL database?

Strong Answer: I start with data access patterns, transaction boundaries, consistency, query flexibility, scale, and operational skills. Relational databases are strong for structured data, joins, constraints, and multi-row transactions. NoSQL stores are useful for flexible aggregates, very high scale, key-based access, or specialized models such as document, graph, time-series, and search.

Selection examples

Requirement	Likely choice
Orders, payments, inventory reservation	Relational database
Session or cache data	Key-value store
Flexible product documents	Document database
Relationship exploration	Graph database
Full-text search and ranking	Search engine

Senior-level point

I do not choose NoSQL simply because the system is large. A relational database with correct indexes, partitioning, read replicas, and archival may be the best design.

11. Data modeling, indexing and partitioning

Interview Question: How do you discuss database design at senior level?

Strong Answer: I model around business invariants and query patterns. I define entity ownership, transaction boundaries, keys, constraints, indexes, history, retention, and hot access paths. I normalize where consistency matters and intentionally denormalize read models where performance and independent evolution justify it.

Senior design checklist

- Create indexes for real query predicates and sort patterns, not every column.
- Use composite index order based on selectivity and access path.
- Plan archival or partitioning before history tables become operational problems.
- Avoid cross-service database joins; expose APIs or build read models.
- Use optimistic concurrency when conflicting updates are possible.

Trade-off

Indexes improve reads but increase write cost, storage, and maintenance. Partitioning improves manageability and pruning, but a poor partition key can create hot partitions or difficult cross-partition queries.

12. Caching and cache invalidation

Interview Question: Where would you use caching, and how do you avoid stale-data issues?

Strong Answer: I cache data only when the latency or downstream load benefit is meaningful and the business can define acceptable staleness. I choose cache-aside, read-through, write-through, or event-driven invalidation based on ownership. Every cache design needs a key strategy, TTL, invalidation rule, size limit, failure behavior, and observability.

Common production risks

Risk	Mitigation
Cache stampede	Request coalescing, jittered TTL, background refresh
Hot key	Replication, sharding, local cache or redesign
Stale data	Short TTL, versioned keys or event invalidation
Cache outage	Graceful database fallback with protection
Sensitive data	Least retention, encryption and access isolation

Good interview line

The cache is normally an optimization, not the source of truth. I design the system to remain correct when the cache is cold or unavailable.

13. Rate limiting and traffic control

Interview Question: How would you design rate limiting for a distributed API?

Strong Answer: I define limits by client, user, tenant, IP, endpoint, and business operation. In a scaled environment, counters need a shared or coordinated store. I select fixed window, sliding window, token bucket, or leaky bucket based on fairness and burst requirements, and return 429 with retry guidance when a limit is exceeded.

Traffic protection layers

- Edge or gateway limits protect the platform from abusive traffic.
- Application limits protect expensive business operations and tenant fairness.
- Concurrency limits protect database connections and downstream dependencies.
- Queue backpressure prevents unlimited backlog growth.
- Circuit breakers stop repeated calls to an unhealthy dependency.

Senior-level point

Rate limits should be configurable, observable, and aligned with commercial or tenant policies. I also distinguish deliberate throttling from overload protection.

14. Authentication, authorization and secrets

Interview Question: What security points do you include in a system design answer?

Strong Answer: I cover identity, token validation, authorization policies, least privilege, secrets management, encryption in transit and at rest, input validation, secure errors, audit logging, data classification, retention, and abuse protection. Authentication proves identity; authorization decides whether that identity can perform the specific action on the specific resource.

Security by layer

Layer	Examples
Edge	TLS, WAF, request limits, token validation
Application	Policy-based authorization, validation, safe errors
Data	Encryption, row or tenant isolation, masking, retention
Operations	Vault-managed secrets, rotation, audit, incident response

Avoid

Do not put secrets in source control, images, logs, or client applications. Do not rely only on UI permissions; backend resources must enforce authorization.

15. Resiliency patterns

Interview Question: Which resiliency patterns do you use for external and internal dependencies?

Strong Answer: I use explicit timeouts, bounded retries with backoff and jitter, circuit breakers, bulkheads, health checks, fallback where it is safe, and graceful degradation. I apply them based on operation semantics. A retry is safe only when the operation is idempotent or protected from duplication.

Pattern and purpose

Pattern	Purpose
Timeout	Stop waiting beyond the business value of the call
Retry	Recover from transient failure without creating a storm
Circuit breaker	Fail fast while a dependency is unhealthy
Bulkhead	Isolate resource pools so one dependency cannot exhaust all capacity
Fallback	Return a safe degraded result when business rules allow it

Senior-level point

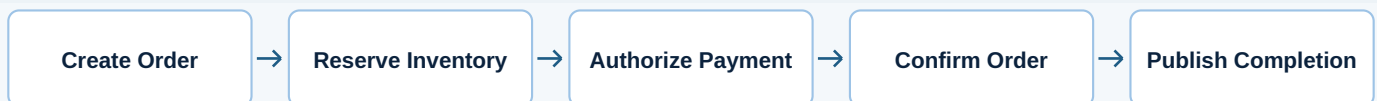
Retries must respect the caller timeout, use a retry budget, and avoid retrying validation errors or known permanent failures.

16. Distributed transactions, Saga and Outbox

Interview Question: How do you maintain consistency when multiple services update their own databases?

Strong Answer: I avoid a large distributed transaction across independently owned services. Each service commits a local transaction, and the workflow progresses through commands and events. A Saga coordinates steps and compensation. A Transactional Outbox atomically stores the business change and event record, and a publisher later sends the event reliably.

Example Saga flow



Failure handling

If payment authorization fails after inventory is reserved, the compensation releases the reservation and the order moves to a failed or cancelled state. Compensation is a business action, not a technical database rollback.

Outbox value

Without an Outbox, the database update can commit while event publication fails, leaving downstream services unaware. The Outbox closes that dual-write gap.

17. Consistency, CAP and reconciliation

Interview Question: How do you decide between strong and eventual consistency?

Strong Answer: I decide based on the business invariant. Strong consistency is required when an incorrect intermediate result can create financial loss, overselling, security exposure, or irreversible action. Eventual consistency is acceptable for notifications, analytics, search indexes, dashboards, and replicated read models when delay is visible and recoverable.

Practical CAP explanation

During a network partition, a distributed system must choose whether a specific operation prioritizes availability or consistency. CAP is not a reason to label an entire system as one type; different workflows can make different choices.

Reconciliation is part of the design

- Persist durable business status and external reference IDs.
- Run scheduled checks for missing, stuck, or conflicting records.
- Compare source-of-truth state with provider or downstream state.
- Repair automatically where safe and route exceptions for manual review.
- Audit every correction.

18. Observability, SLI/SLO and alerting

Interview Question: How do you make a system supportable in production?

Strong Answer: I design structured logs, correlation and trace IDs, metrics, distributed traces, health checks, dashboards, alerts, and audit events from the beginning. I define service-level indicators such as latency, error rate, availability, throughput, and backlog, then set objectives that reflect the user experience and business criticality.

Useful signals for a workflow system

Signal	Examples
User experience	p50/p95/p99 latency, success rate
Work queues	Depth, oldest message age, retry and DLQ count
Dependencies	Timeouts, circuit state, provider error rate
Business	Jobs completed, payments reconciled, imports failed
Deployment	Version, instance health, error change after release

Senior-level point

An alert should indicate a user-impacting or actionable condition. Alerting on every exception creates noise and delays real incident response.

19. Case study: large file import processing

Interview Question: How would you design a system that imports files containing millions of records?

Strong Answer: I separate upload, durable job creation, validation, transformation, batch processing, status tracking, error handling, and notification. The API quickly returns a job ID. Background workers process bounded batches, persist checkpoints, and make each step retryable and idempotent. The user tracks progress through a status API instead of holding an HTTP request open.

High-level flow



Core data model

Entity	Important fields
ImportJob	JobId, tenant, file URI, status, total, processed, failed, created, completed
ImportBatch	BatchId, range, attempt, status, checkpoint, worker lease
ImportError	JobId, row, code, safe message, raw-value reference
StatusHistory	From status, to status, timestamp, actor, correlation ID

Failure and scale considerations

- Stream or stage the file; do not load the entire file into application memory.
- Use bounded batch sizes and worker concurrency to protect the database.
- Persist checkpoints so a worker restart does not repeat the entire import.
- Use idempotent upsert or business keys to prevent duplicate records.
- Move unrecoverable batches to a review queue while preserving partial results if business rules allow.
- Expose progress, warnings, downloadable error details, cancellation rules, and audit history.

Real project answer

In an operational planning platform, I would store the uploaded file outside the web application, create a durable JobQueue record, and let multiple workers claim batches using leases. SQL indexes support job status and date searches, while dashboards track throughput, backlog age, retry count, and the failed processing step.

20. Case study: notification service

Interview Question: How would you design a centralized email, SMS, and push notification service?

Strong Answer: The service accepts a request or event, validates recipient and template data, applies tenant and preference rules, renders the message, sends through a channel provider, and records delivery state. Core business workflows publish notification intent asynchronously so provider failures do not block the transaction.

Core capabilities

- Template versioning, localization, channel preference, suppression and unsubscribe rules.
- Per-tenant and per-provider throttling, retries with backoff, DLQ and replay.
- Provider abstraction so channels can fail over or change independently.
- Delivery status, correlation ID, audit trail, dashboards and cost monitoring.

Senior-level trade-off

For urgent security messages I may try an immediate primary channel and then fall back. For routine workflow emails, I prioritize durability and eventual delivery over synchronous response time.

21. Case study: payment or order workflow

Interview Question: How would you design a reliable order and payment workflow?

Strong Answer: I create an order with a durable state machine, validate the request, reserve inventory, authorize payment using idempotency, and confirm or compensate based on each result. Provider callbacks are authenticated and deduplicated. Reconciliation jobs verify that internal state matches the payment provider and fulfillment systems.

Example state transitions

Created -> Validated -> InventoryReserved -> PaymentAuthorized -> Confirmed. Failure states are explicit, and every transition records who or what caused it.

Senior-level points

- Never infer final payment state only from a client response.
- Use provider reference IDs, idempotency keys, immutable audit events, and reconciliation.
- Protect inventory from oversell with an atomic reservation strategy.
- Design manual review for unresolved or conflicting outcomes.

22. Background job scheduling

Interview Question: How do you design scheduled and long-running background jobs?

Strong Answer: I separate scheduling from execution. The scheduler creates durable job instances; workers claim jobs with a lease, execute idempotently, update heartbeat and progress, and release or retry on failure. I prevent overlapping executions when the business process requires a singleton.

Production controls

- Distributed lock or lease with expiry, not an in-memory flag.
- Misfire policy for jobs missed during downtime.
- Maximum runtime, cancellation, retry limit, and dead-letter state.
- Job version, input snapshot, correlation ID, owner and run history.
- Dashboard for next run, active runs, stale heartbeats and failures.

Real project lens

For an ETL process triggered by SQL Agent or a scheduler, I would insert a JobQueue record and let a Windows Service or worker claim it. The scheduler does not perform the data processing itself.

23. Audit trail and compliance

Interview Question: How would you design an audit trail for sensitive business operations?

Strong Answer: I capture who performed the action, what changed, when, where the request originated, the business reason, correlation ID, and the before/after values permitted by policy. Audit records are append-only, access-controlled, retained according to regulation, and protected from secret or sensitive-data leakage.

Audit is not the same as application logging

Application logs help diagnose software behavior. Audit records provide durable business evidence. They have stricter integrity, retention, access, and reporting requirements.

Senior-level points

- Use UTC timestamps and stable actor identities.
- Mask or tokenize sensitive fields; do not copy secrets into audit data.
- Record administrative access and audit-report access as well.
- Define retention, legal hold, export, and tamper-detection requirements.

24. Case study: multi-tenant SaaS design

Interview Question: How would you design a SaaS platform that serves many customers while preserving tenant isolation?

Strong Answer: I make tenant identity a first-class part of authentication, authorization, data access, caching, messaging, observability, and rate limiting. I choose a data-isolation model based on compliance, scale, noisy-neighbor risk, operational cost, and customer requirements. Every request receives a trusted tenant context from validated identity, never from an unverified client field.

Data isolation models

Model	Advantages	Trade-offs
Shared database, shared schema	Lowest cost and easiest aggregate operations	Requires strict tenant predicates and controls
Shared database, schema per tenant	More logical isolation	Schema lifecycle and connection complexity
Database per tenant	Strong isolation and customer-specific operations	Higher cost, provisioning and fleet management

End-to-end isolation checklist

- Tenant-aware authorization policies and row-level or repository-level filtering.
- Tenant included in cache keys, message headers, partition keys, storage paths, metrics, and audit records.
- Per-tenant quotas, rate limits, concurrency controls, and cost visibility.
- Tenant-safe background jobs: the worker restores tenant context before accessing data.
- Automated tests that attempt cross-tenant reads and writes.
- Provisioning, migration, backup, restore, export, deletion, and offboarding workflows.

Noisy-neighbor protection

I combine per-tenant throttles, queue partitions, worker concurrency limits, database resource controls, and usage dashboards. Premium or regulated tenants may receive dedicated resources when the business case requires stronger isolation.

25. High availability and disaster recovery

Interview Question: How do you design for availability, regional failure, and data recovery?

Strong Answer: I begin with business-defined recovery objectives. RTO is how quickly service must return; RPO is how much data loss is acceptable. I remove single points of failure within the required fault domain, use health-based routing, test backups and failover, and document how the application behaves during partial dependency failure.

Design by failure domain

Failure	Design response
Single instance	Multiple instances, load balancer, health probes
Availability zone	Zone-redundant compute and data where required
Database primary	Replica or managed failover plus tested connection behavior
Region	Warm standby or active-active based on RTO/RPO and cost
Operator error or corruption	Point-in-time restore, immutable backup, controlled recovery

Senior-level trade-offs

- Active-active improves availability but makes data consistency, routing, and conflict handling harder.
- A backup is not a recovery strategy until restore time and application correctness are tested.
- Health checks must distinguish liveness, readiness, and dependency degradation.
- Failover automation needs guardrails to avoid switching to an equally unhealthy target.
- Run game days and capture recovery steps, owners, communication, and rollback criteria.

Good interview line

"I would not promise zero downtime or zero data loss without validating the business requirement, database replication model, consistency impact, and operational cost."

26. Deployment, scaling and production readiness

Interview Question: What do you include to make the design deployable and operable in the cloud?

Strong Answer: I keep application instances stateless where practical, externalize configuration and secrets, expose health endpoints, use immutable artifacts, automate tests and deployment gates, and design horizontal scaling around real bottlenecks. I explain rollout, rollback, schema compatibility, observability, and capacity protection.

Release flow



Production-readiness checklist

- Backward-compatible API and database changes support mixed application versions during rollout.
- Readiness fails when an instance should not receive traffic; liveness restarts only truly stuck processes.
- Autoscaling uses relevant signals such as CPU, request concurrency, latency, or queue depth.
- Resource limits, connection pools, thread pools, and downstream quotas are aligned.
- Blue-green, canary, or rolling deployment is selected based on risk and platform capability.
- Rollback includes database and message-contract strategy, not only application binaries.
- Dashboards compare the new version with the previous baseline before full rollout.

How developers usually explain CI/CD

As a developer, I push code through the agreed CI/CD workflow. The pipeline builds, tests, scans, packages the application into an immutable artifact or container image, promotes it through environments, and deploys to the target service. I understand the stages and deployment behavior even when a platform team owns the pipeline configuration.

27. Final senior system design answer framework

Interview Question: How should you structure a 30 to 45 minute system design answer?

Strong Answer: I use a repeatable structure so the conversation remains clear. I allocate time to requirements, scale, architecture, APIs and data, one or two deep dives, failure handling, security, operations, and trade-offs. I continuously confirm that I am solving the interviewer's intended problem.

Suggested interview timeline

Time	Focus	What to say
0-5 min	Clarify	Users, workflows, constraints, assumptions, success criteria
5-10 min	Estimate	Peak load, data growth, latency, availability, consistency
10-20 min	High-level design	Request flow, services, stores, messaging, ownership
20-32 min	Deep dive	APIs, schema, concurrency, scaling, critical workflow
32-40 min	Production	Failure handling, security, observability, deployment
40-45 min	Close	Trade-offs, bottlenecks, risks and next improvements

Senior phrases you can use naturally

- "I am making this assumption because the requirement is not yet specified."
- "This component is the source of truth; the read model can be eventually consistent."
- "I would start simpler and introduce this complexity only when the scale or ownership requires it."
- "The retry is safe because the operation is idempotent and protected by a unique key."
- "The main trade-off is lower latency versus staleness and invalidation complexity."
- "I also want to cover how support teams detect, diagnose, and recover from failure."

Common follow-up questions to practice

- What becomes the first bottleneck at 10x scale?
- How do you prevent duplicate processing and out-of-order updates?
- What happens if the database, cache, broker, or external provider is unavailable?
- How do you deploy a breaking schema change without downtime?
- How do you isolate one tenant or client that overloads the system?
- Which metric proves the user experience is healthy?

Quick Revision Sheet

Concept	Interview focus
Approach	Clarify -> estimate -> design -> deep dive -> production -> trade-offs
Requirements	Functional, NFR, scale, SLA, security, failure and assumptions
Architecture	Request flow, ownership, sync/async boundaries and source of truth
Services	Start simple; split by business capability and ownership
APIs	Stable contracts, validation, versioning, pagination and idempotency
Messaging	At-least-once mindset, idempotent consumer, ordering, DLQ and Outbox
Data	Access patterns, transactions, indexes, partitioning, retention and concurrency
Caching	Latency benefit versus staleness, invalidation and stampede risk
Security	Identity, resource authorization, secrets, encryption, audit and limits
Resiliency	Timeout, bounded retry, circuit breaker, bulkhead and degradation
Consistency	Business invariant, Saga, compensation, eventual consistency and reconciliation
Observability	Logs, metrics, traces, business signals, SLI/SLO and actionable alerts
Case studies	Durable state, explicit workflow, failure recovery and operational controls
Cloud readiness	Stateless scale, health, immutable deployment, compatibility and rollback
Closing answer	State trade-offs, bottlenecks, risks, alternatives and next steps

THANK YOU

For Supporting This Series

C# & .NET Deep Dive - Interview Prep Parts 1 to 8

Your encouragement, feedback, shares, and support made this series possible.

I created these guides to help developers prepare with clarity, connect concepts to real production scenarios, and answer senior-level interview questions with confidence.

Keep learning. Keep building. Keep helping others grow.

With gratitude,

Anusha Gummula

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